

Feature ID	Feature Name	Feature Description	Comment
1	request_id	request id	not trainable
2	item_id	item id	not trainable
3	itemimps	weighted number of item's impressions based on position	not trainable
4	item_shows	number of item's impressions	not trainable
5-8	bm_*	bm25 based	
9-14	mcat*	categories	categorical
15	cat_id	subcategory id	categorical
16-19	tfidf_*	tf-idf based	
20-30	idf_*	idf based	
31-36	trf_*	trigramm based	
37-40	query_info_*	query text characteristic	
41-45	item_info	item text characteristic	
46	same_loc	indicator user&item in the same location $\in \{0, 1\}$	
47	dist	the distance from the item to the user	
48	item_delivery	item's delivery indicator $\in \{0, 1\}$	
49	item_region_id	item region id	categorical
50	item_location_id	item location id	categorical
51-55	word_count_*	based on the number of unique request tokens	
56-58	hit_count_*	based on the non-unique token matches	
59-62	hit_pos_*	positional features based on matches	
63-65	len_*	based on length of text	
66-70	lcs_*	longest common sequence	
71-74	wlccs_*	weighted longest common contiguous subsequence	
75-78	lccs_*	longest common contiguous subsequence	
79-81	atc_*	aggregate term closeness	
82-87	phrase_decay_*	based on phrase decay	
88-90	group_ctr_*	based on mean item's group ctr	
91	ctc	conversion (click to contact)	
92	ctr	conversion (impression to click)	x^b for UCB-FE
93	target	click indicator $\in \{0, 1\}$	target

Table 1: Feature Description

The train and test datasets together account for approximately 16_000_000 logged production data, derived from over 530_000 user queries (*request_id*). The data contains more than 9_000_000 unique items(*item_id*).